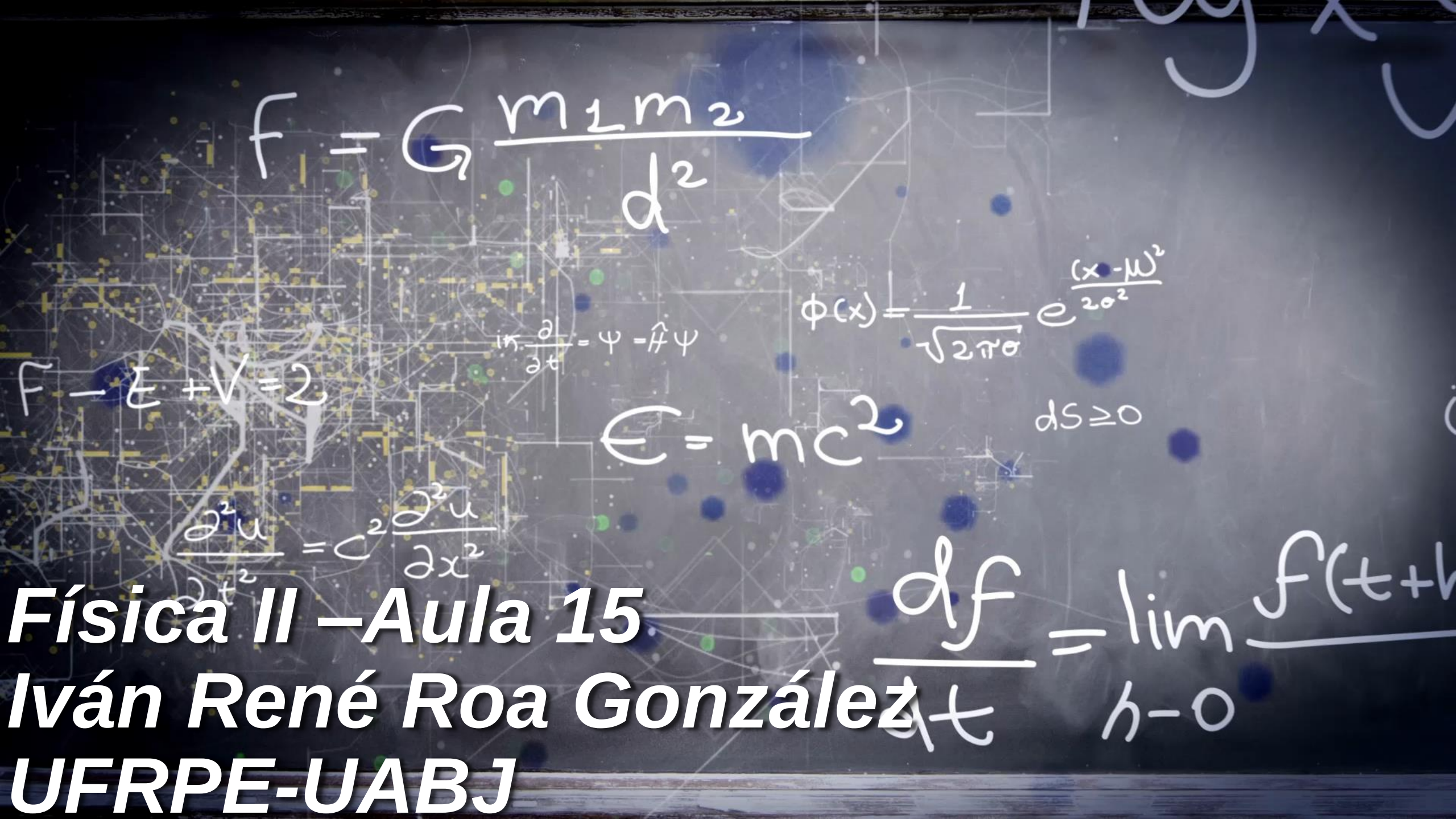
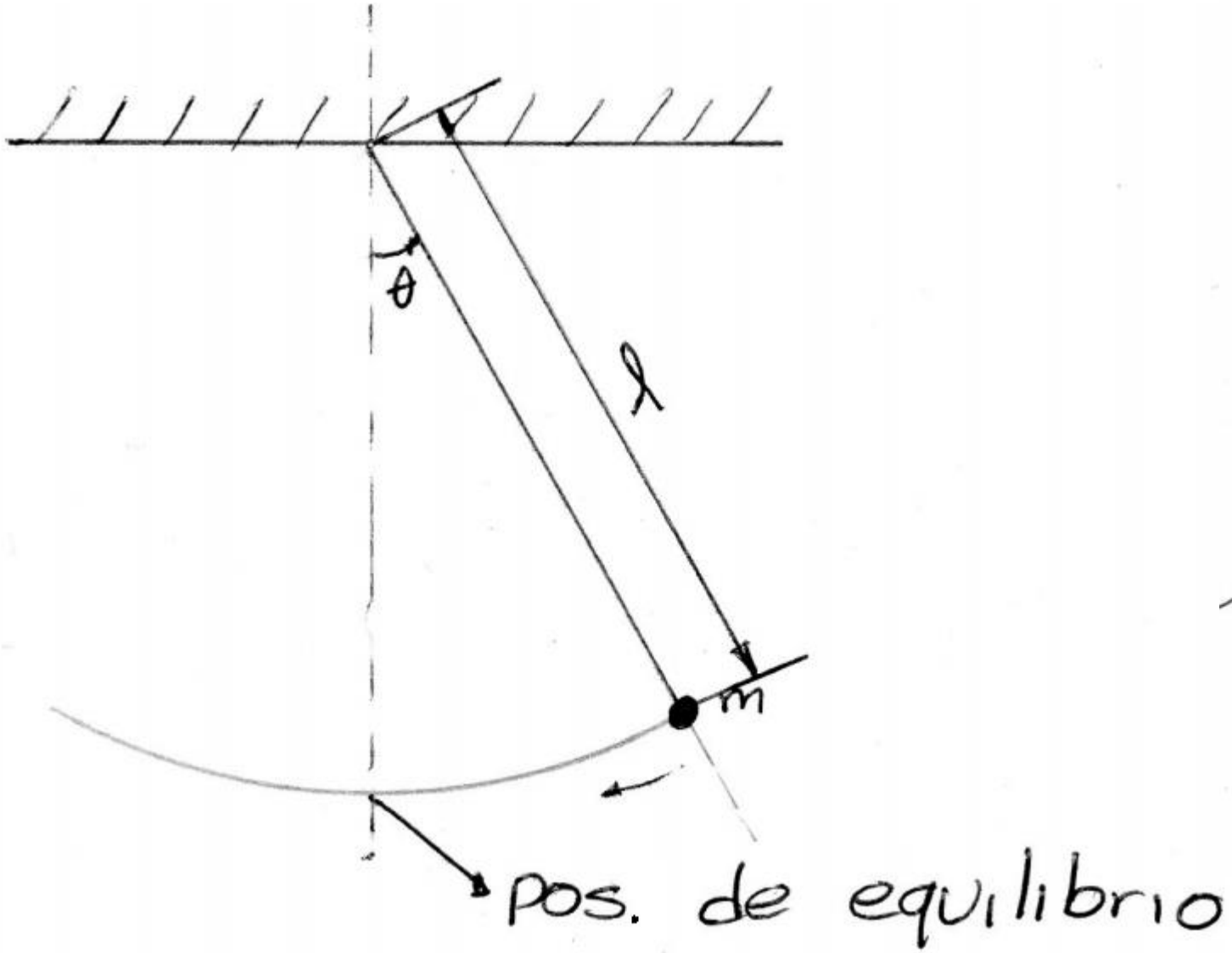


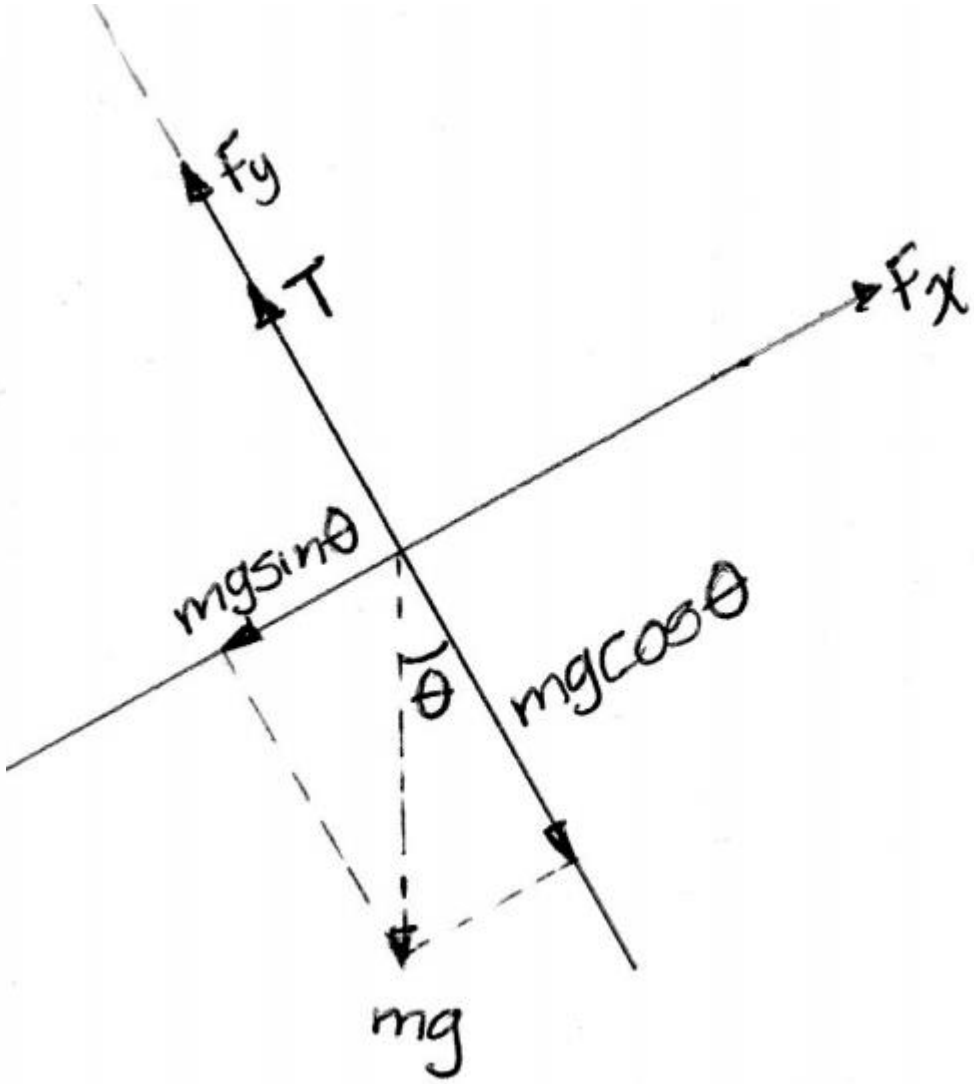
Física II –Aula 15

Iván René Roa González

UFRPE-UABJ

Pêndulo simples



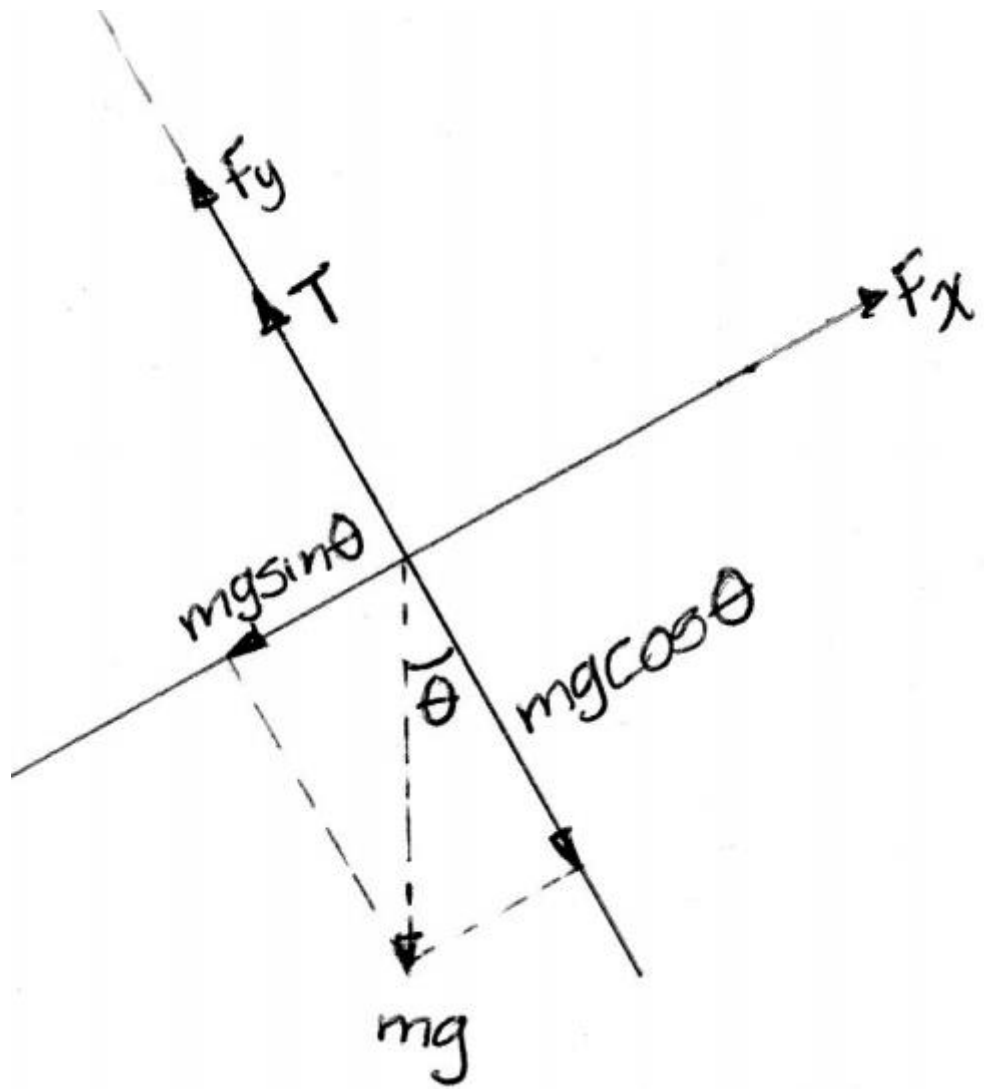


$$\sum F_y = T - mg\cos\theta = 0$$

$$T = mg\cos\theta$$

$$\sum F_x = -mg\sin\theta = ma_x$$

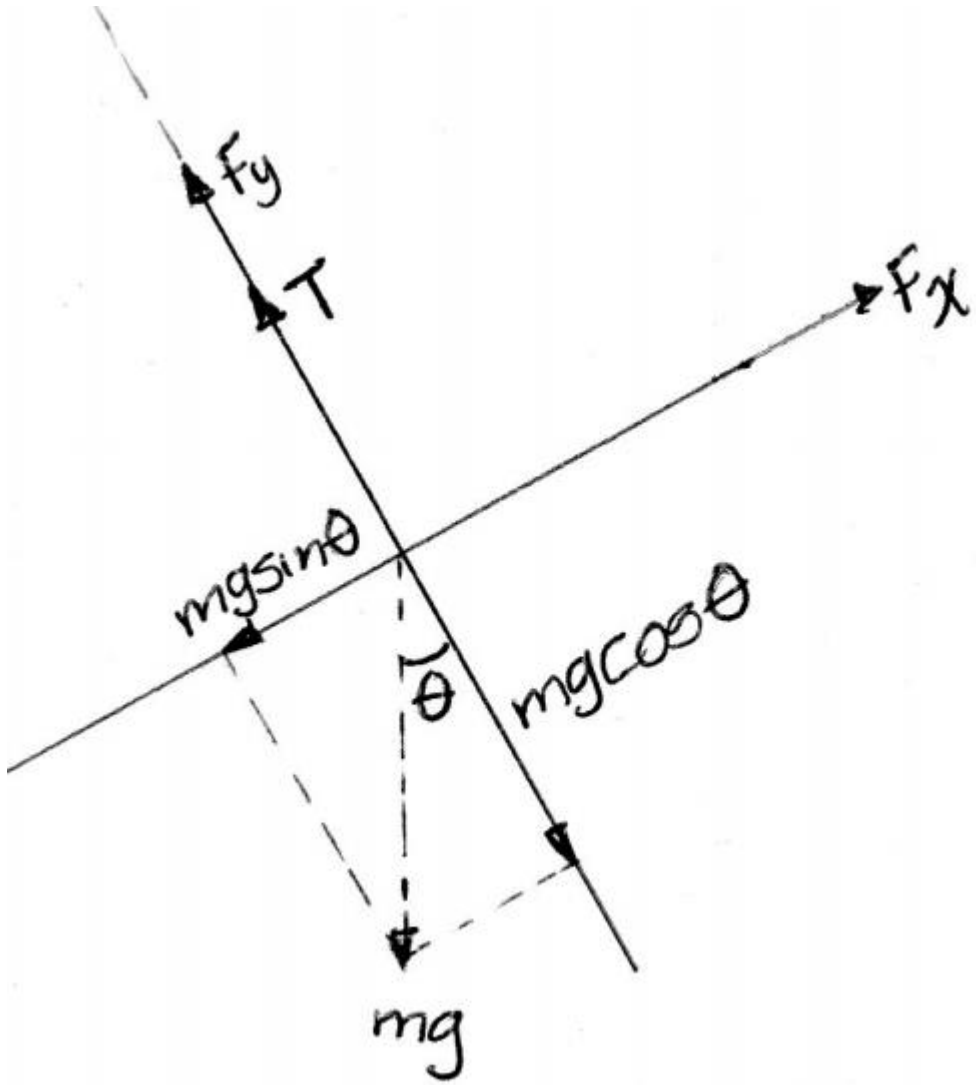
$$a_x = l\frac{d^2\theta}{dt^2}$$



$$-\cancel{m}g\sin\theta = \cancel{m}\,l\frac{d^2\theta}{dt^2}$$

$$\frac{d^2\theta}{dt^2} = -\frac{g}{l}\sin\theta$$

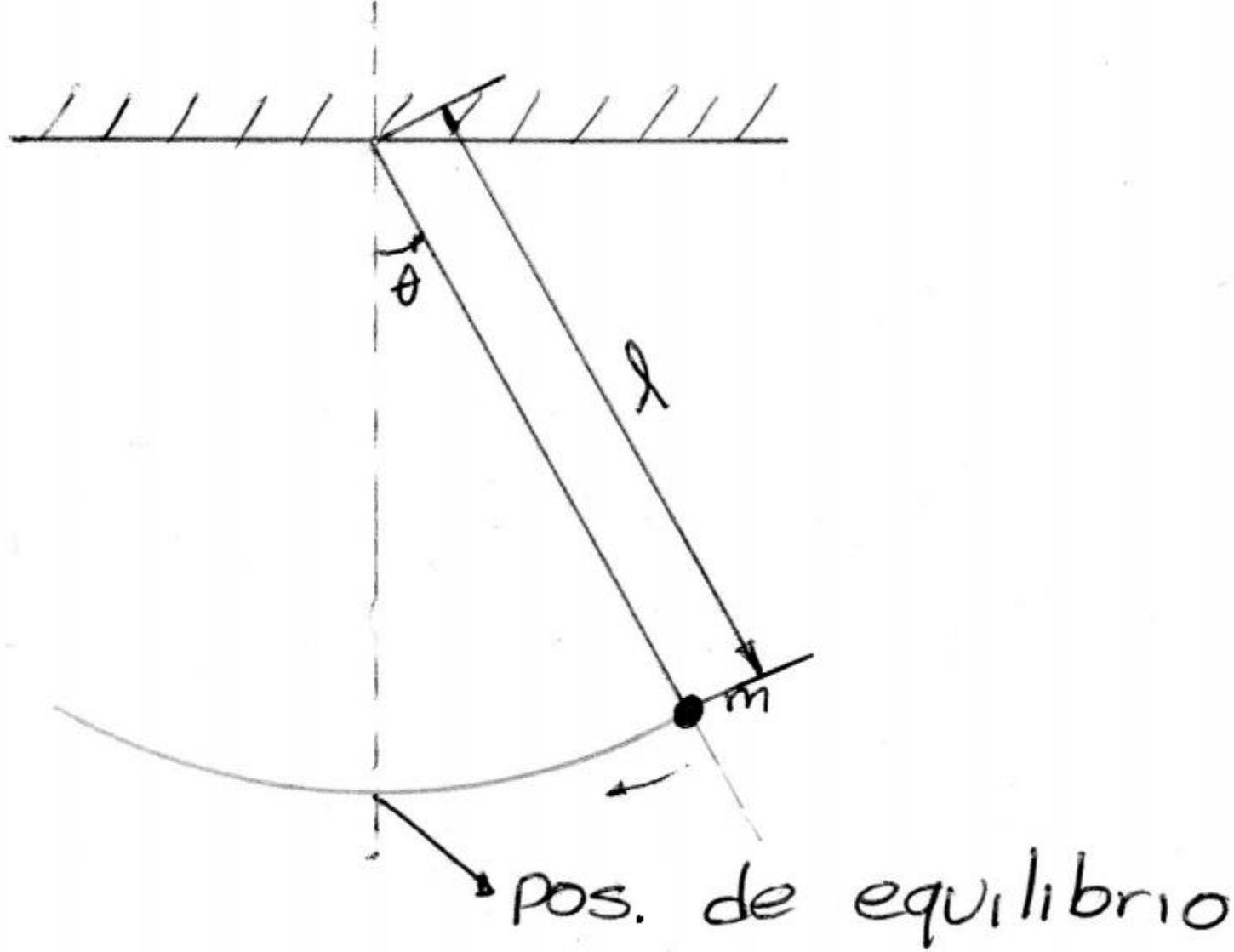
$$\frac{d^2\theta}{dt^2} + \frac{g}{l}\sin\theta = 0$$



$\theta \ll 1$ rad

$0 < \theta < 20°$

$f(x) = \sum a_n x^n$



$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \dots + \frac{f^{(n)}}{n!}x^n + \dots$$

$$\sin\theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \dots + (-1)^n \frac{\theta^{2n+1}}{(2n+1)!} + \dots$$

$$\theta \ll 1\,\text{rad}$$

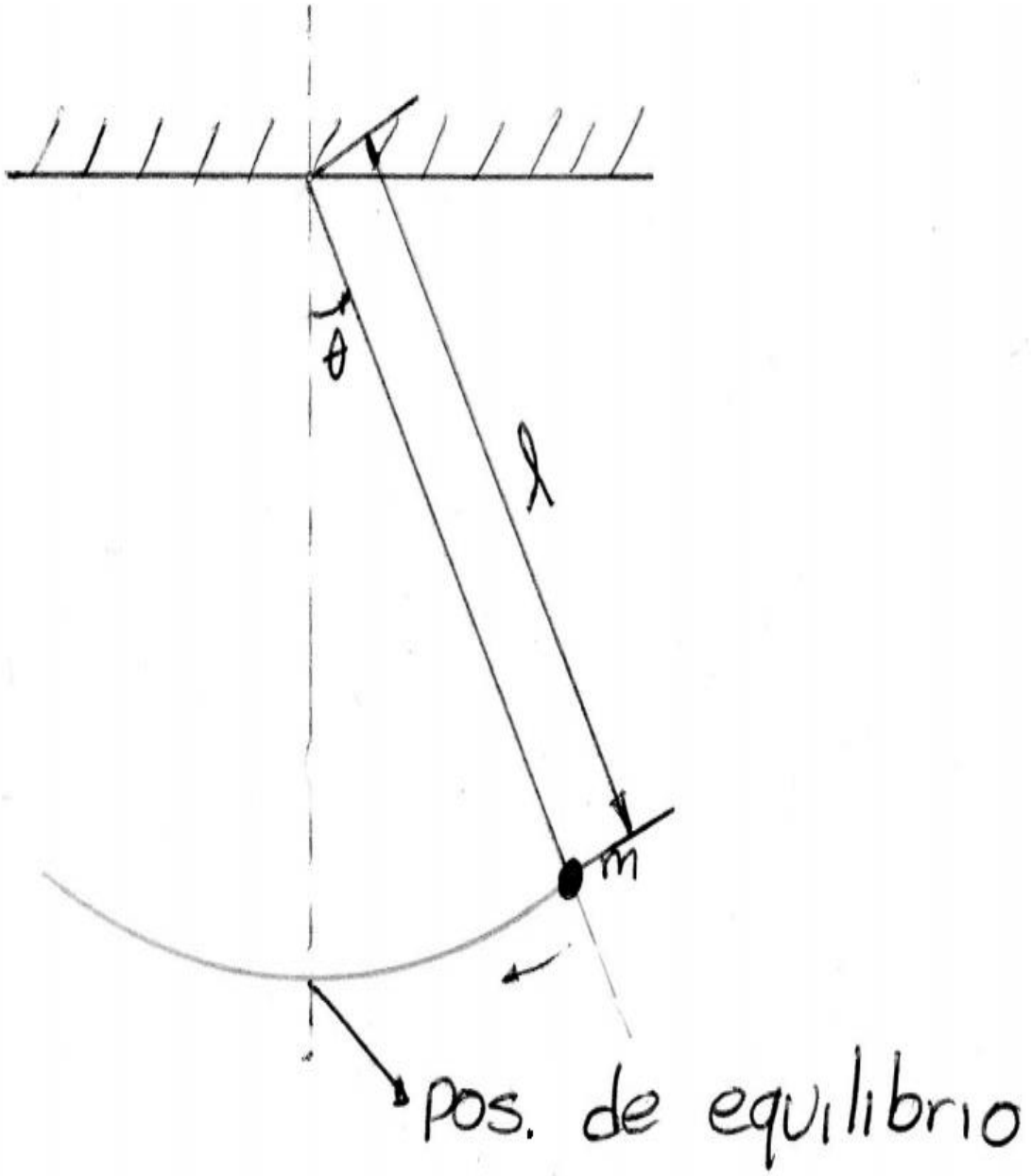
$$\sin\theta \approx \theta$$

$$\frac{d^2\theta}{dt^2} + \frac{g}{l}\theta = 0$$

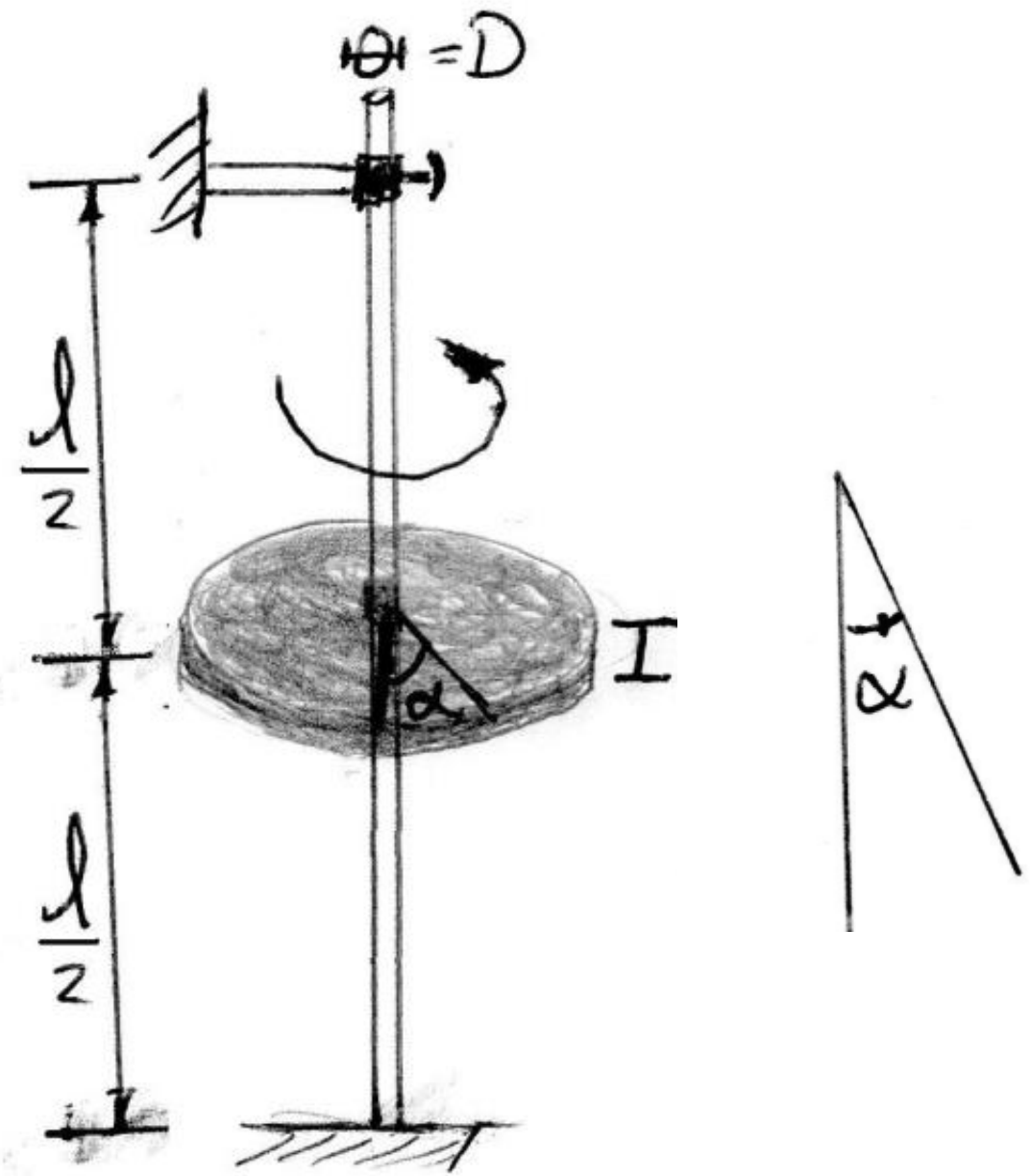
$$\omega_0^2 = \frac{[F]}{[M][L]} \quad ; \quad \omega_0^2 = \frac{\cancel{m}g}{\cancel{m}l} = \frac{g}{l}$$

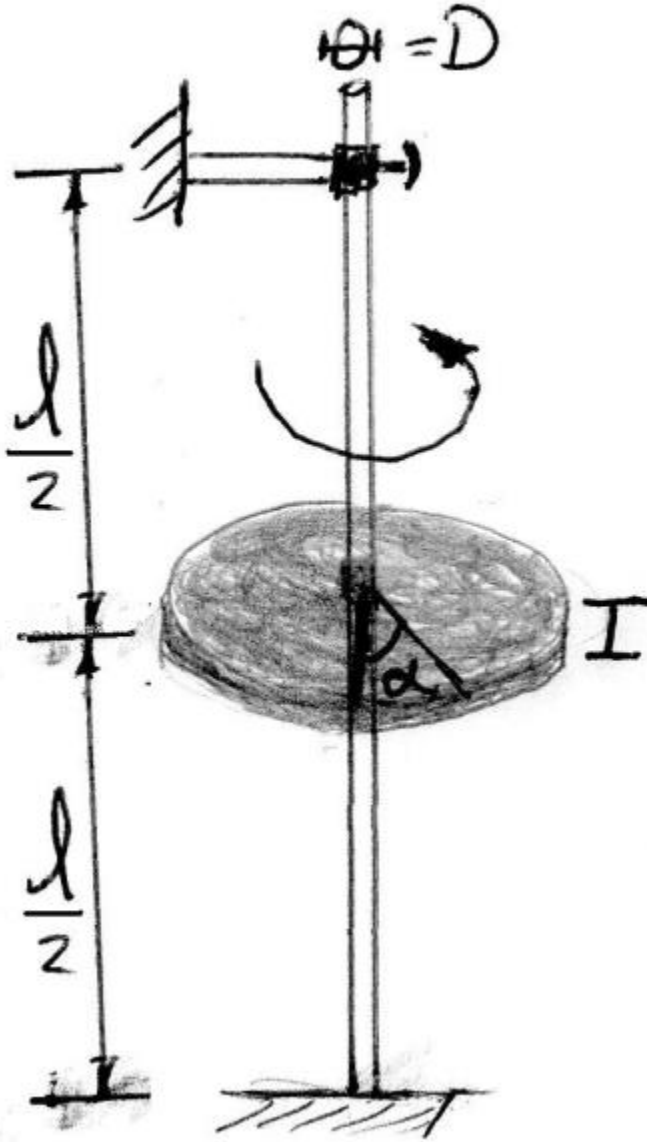
$$\theta(t) = \theta_0 \sin(\omega_0 t + \varphi)$$

$$\boxed{\theta(t) = \theta_0 \sin\left(\sqrt{g/l}\, t + \varphi\right)}$$



Pêndulo de torção



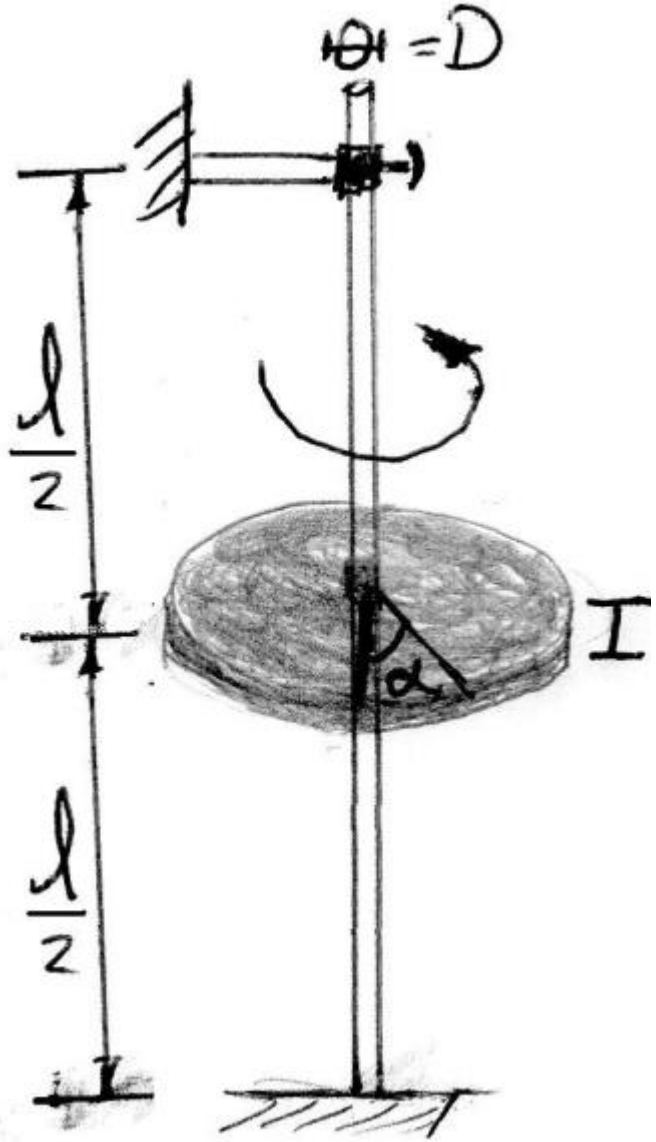


$$K = K(G, D, l)$$

$$\sum \vec{\tau} = I\vec{\ddot{\alpha}} \quad ; \quad \ddot{\alpha} = \frac{d^2\alpha}{dt^2}$$

$$-K\alpha = I\frac{d^2\alpha}{dt^2}$$

$$\frac{d^2\alpha}{dt^2} + \frac{K}{I}\alpha = 0$$



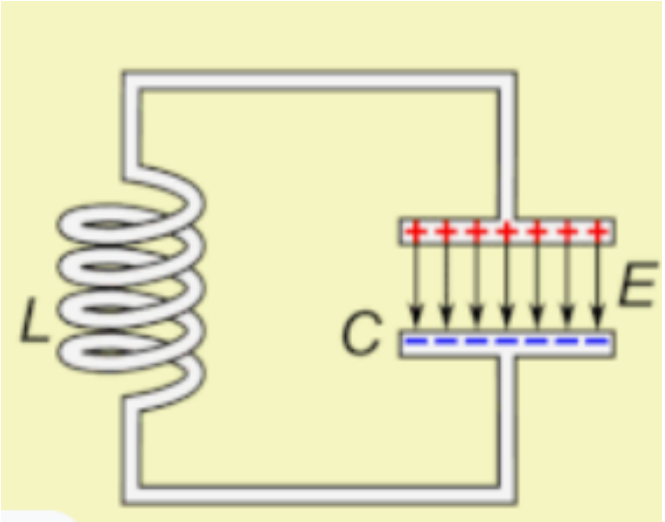
$$\alpha(t) = \alpha_0 \sin(\omega_0 t + \varphi)$$

$$\alpha(t) = \alpha_0 \sin(\sqrt{K/I}\, t + \varphi)$$

$$\omega_0^2 = \frac{[\tau]}{[I][\theta]}$$

Circuito LC
Segunda lei de Kirchhoff

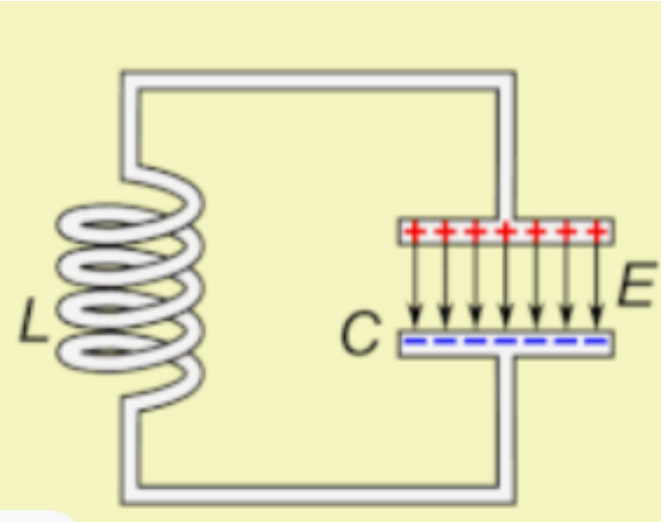
- A segunda Lei de Kirchhoff afirma que ao percorrer uma malha por um determinado sentido, partindo e chegando ao mesmo ponto, a soma algébrica das d.d.p. é nula, ou seja, a soma das tensões em uma malha deve ser igual a zero.



$$\sum \Delta V = 0$$

$$\Delta V = \frac{1}{C} q$$

$$\Delta V = L \frac{dI}{dt} \longrightarrow \Delta V = L \frac{d^2 q}{dt^2}$$

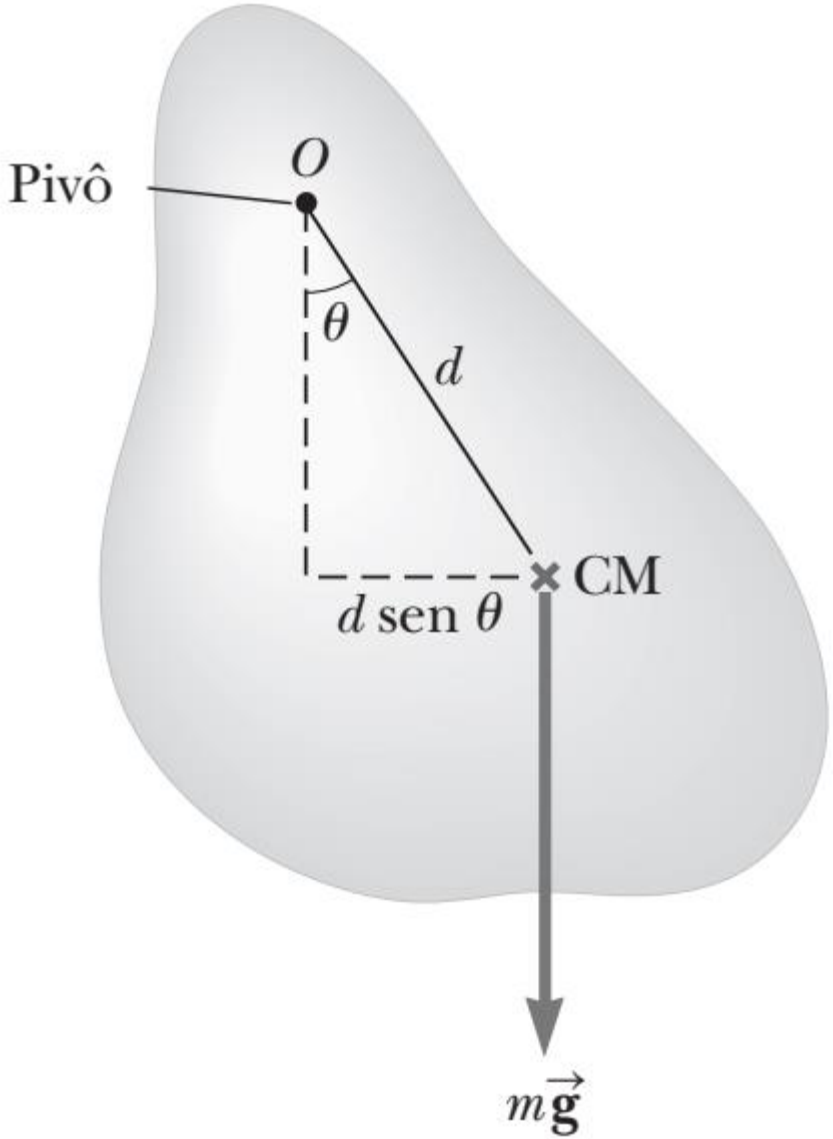


$$L \frac{d^2 q}{dt^2} + \frac{1}{C} q = 0$$

$$\frac{d^2 q}{dt^2} + \frac{1}{LC} q = 0 \quad \Rightarrow \quad \omega_0 = \sqrt{\frac{1}{LC}}$$

$$\boxed{q(t) = q_0 \sin(\omega_0 t + \varphi)}$$

O pêndulo físico



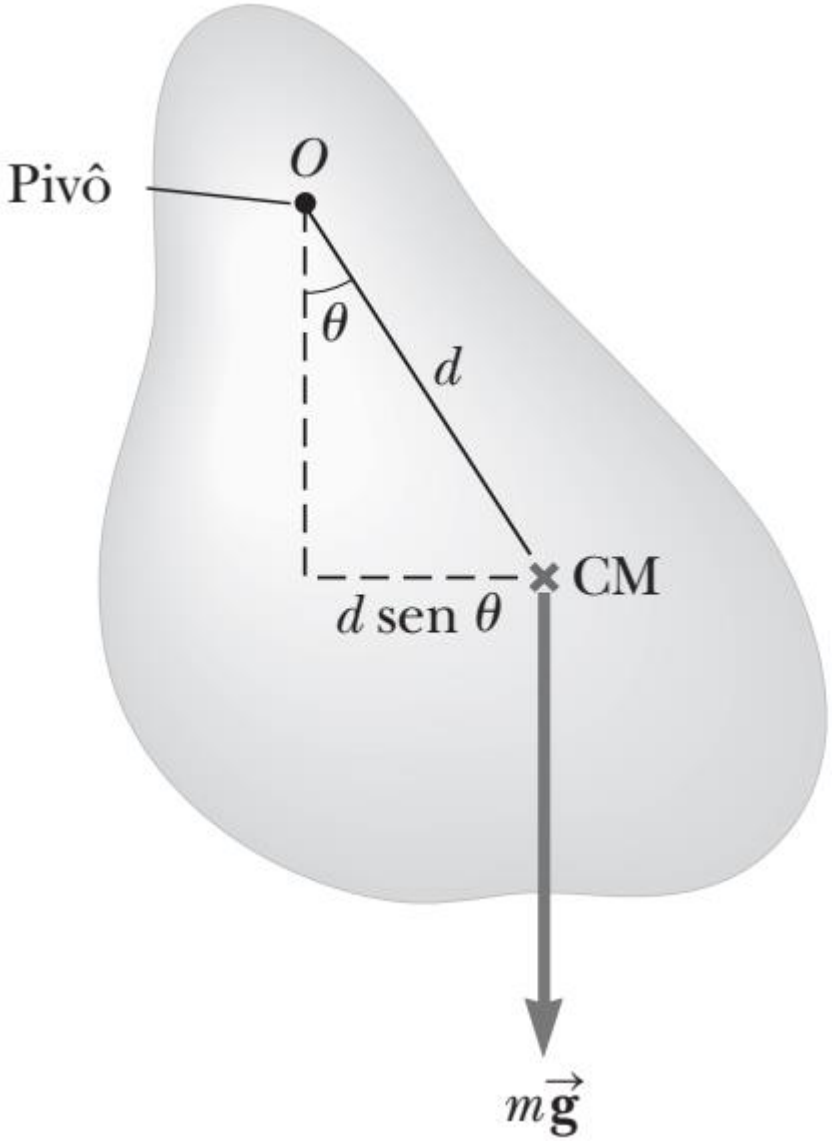
$$\Sigma\tau_{\text{ext}} = I\alpha$$

$$-mgd\,\text{sen}\,\theta = I\frac{d^2\theta}{dt^2}$$

$$\text{sen}\,\theta \approx \theta$$

$$\frac{d^2\theta}{dt^2} = -\left(\frac{mgd}{I}\right)\theta = -\omega^2\theta$$

O pêndulo físico

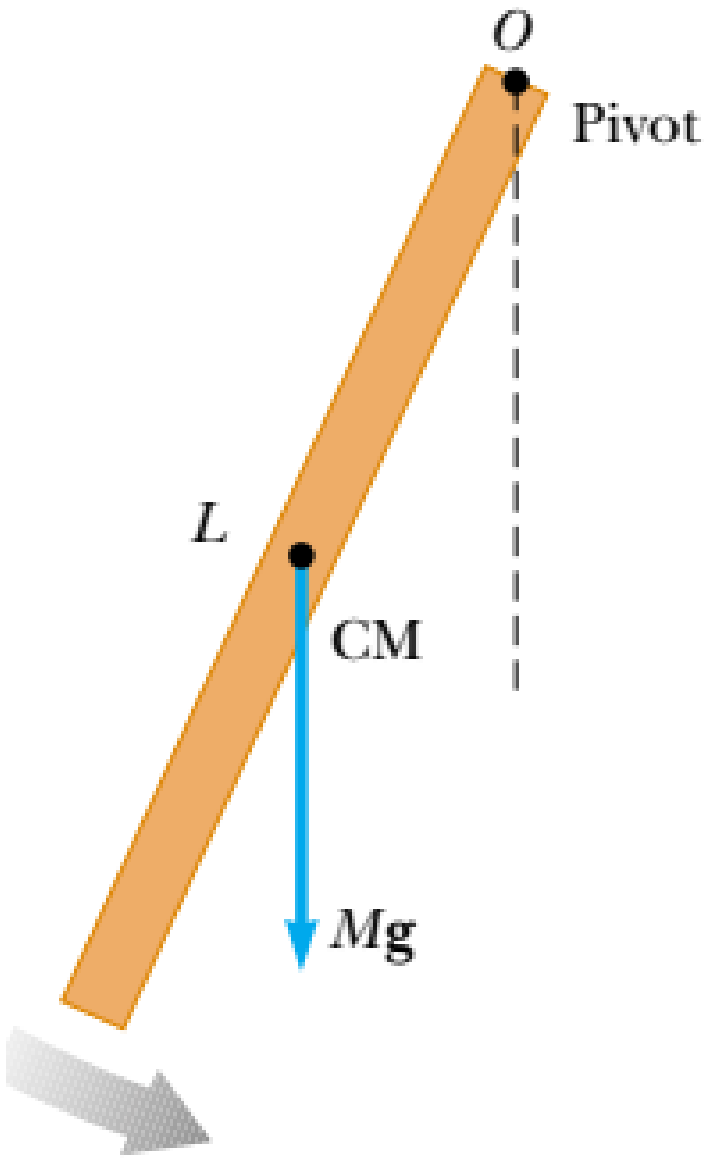


$$\theta = \theta_{\text{máx}} \cos(\omega t + \phi)$$

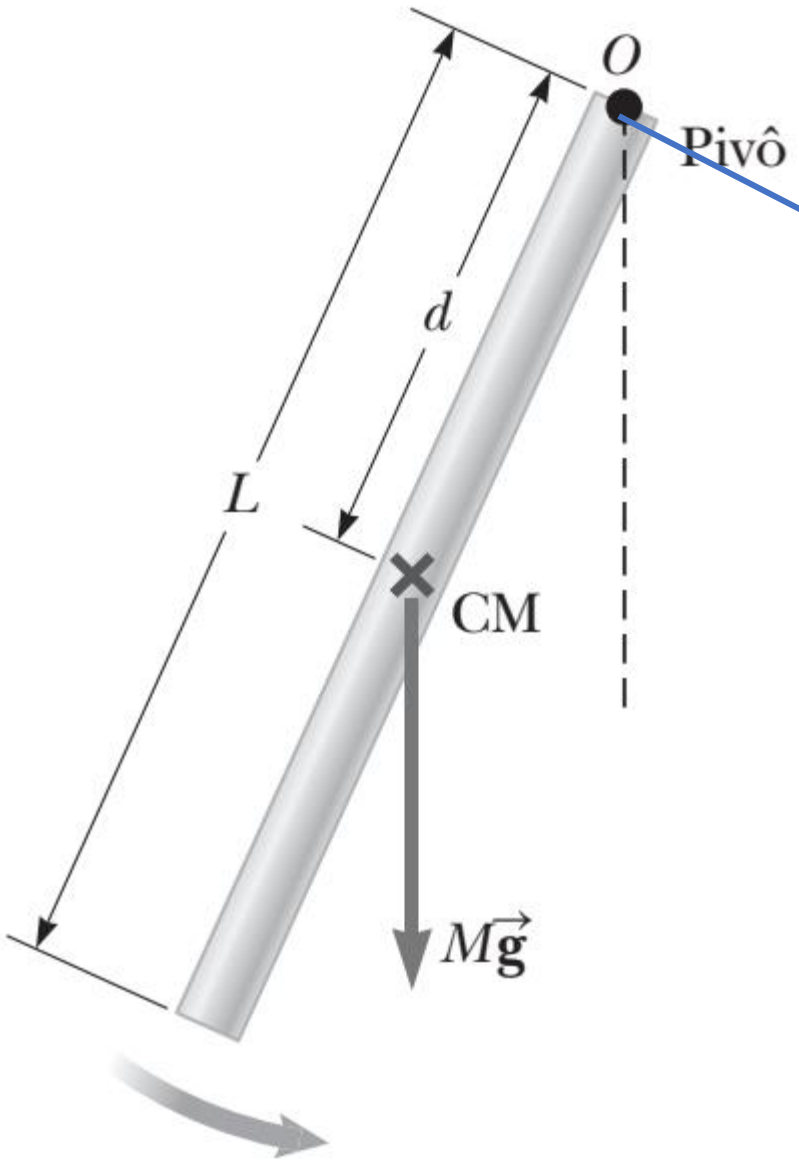
$$\omega = \sqrt{\frac{mgd}{I}}$$

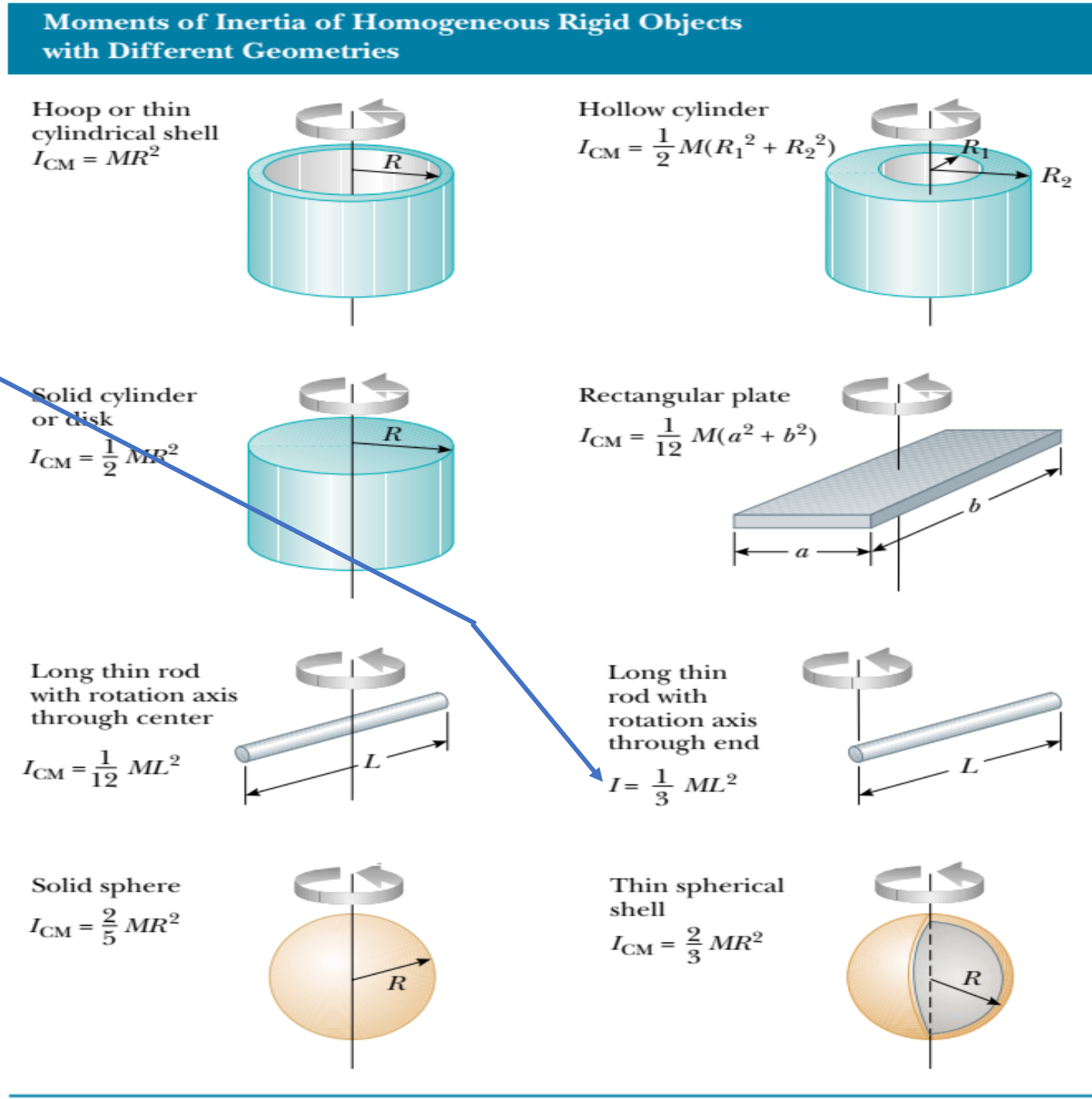
$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{I}{mgd}}$$

Uma haste oscilante



Uma haste uniforme de massa M e comprimento L é suspensa por um pivô em uma extremidade e oscila em um plano vertical (Fig.). Encontre o período de oscilação se a amplitude do movimento for pequena.


O
Pivô
d
L
CM
$M\vec{g}$


Moments of Inertia of Homogeneous Rigid Objects with Different Geometries
Hoop or thin cylindrical shell
$I_{CM} = MR^2$
R
Hollow cylinder
$I_{CM} = \frac{1}{2}M(R_1^2 + R_2^2)$
R_1
R_2
Solid cylinder or disk
$I_{CM} = \frac{1}{2}MR^2$
R
Rectangular plate
$I_{CM} = \frac{1}{12}M(a^2 + b^2)$
a
b
Long thin rod with rotation axis through center
$I_{CM} = \frac{1}{12}ML^2$
L
Long thin rod with rotation axis through end
$I = \frac{1}{3}ML^2$
L
Solid sphere
$I_{CM} = \frac{2}{5}MR^2$
R
Thin spherical shell
$I_{CM} = \frac{2}{3}MR^2$
R

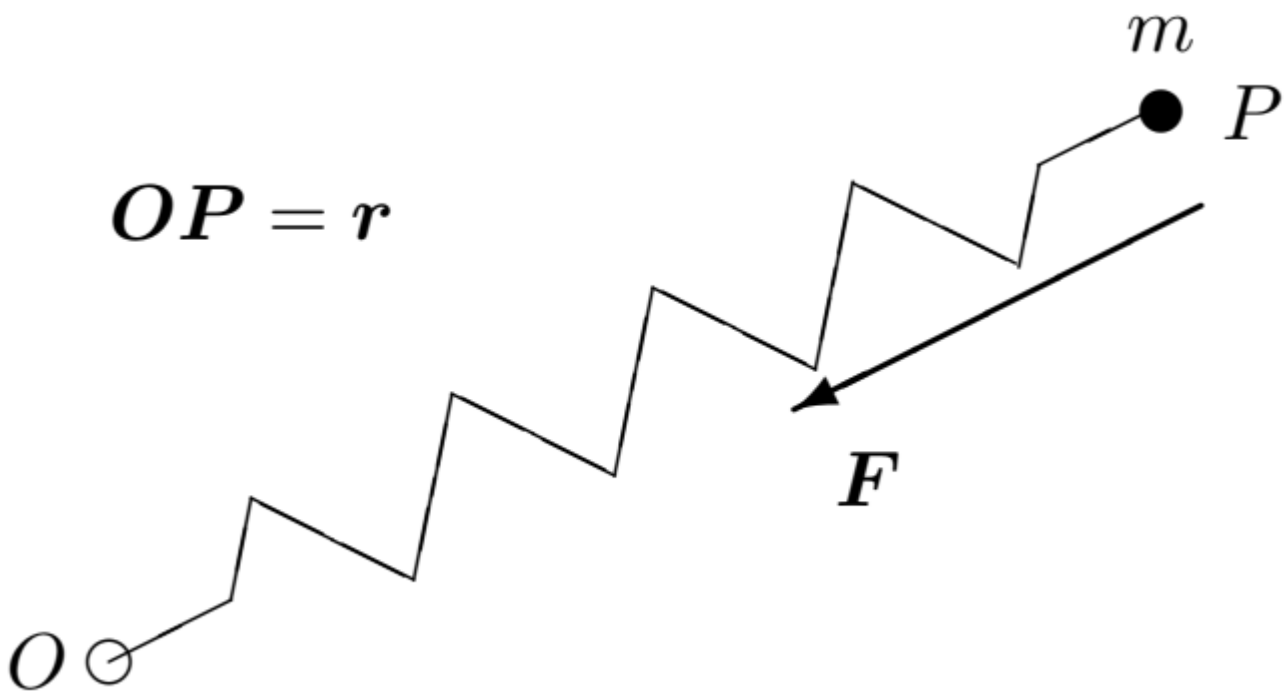
Conceitualização Imagine uma haste balançando para a frente e para trás quando suspensa por uma extremidade. Tente fazer isso com uma régua de metro ou um pedaço de madeira.

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{I}{mgd}}$$

$$T = 2\pi\sqrt{\frac{\frac{1}{3}ML^2}{Mg(L/2)}} = 2\pi\sqrt{\frac{2L}{3g}}$$

Oscilações em 2 dimensões

- Seja uma massa m que pode ser movida em um plano, atraída para um ponto O do mesmo por uma mola linear.



$$\boldsymbol{F} = -k\boldsymbol{r}$$

$$m\ddot{\boldsymbol{r}} + k\boldsymbol{r} = \boldsymbol{0}$$

$$m\ddot{\boldsymbol{r}} + k\boldsymbol{r} = \boldsymbol{0}$$

$$m\ddot{x} + kx = 0; \qquad m\ddot{y} + ky = 0.$$

$$x = A\cos(\omega_0 t + \alpha) \qquad y = B\cos(\omega_0 t + \beta)$$

$$\alpha = 0; \; \delta = \beta - \alpha$$

$$y = B(\cos\omega_0 t \cos\delta - \operatorname{sen}\omega_0 t \operatorname{sen}\delta)$$

$$y = B(\cos\omega_0 t \cos\delta - \operatorname{sen}\omega_0 t \operatorname{sen}\delta),$$

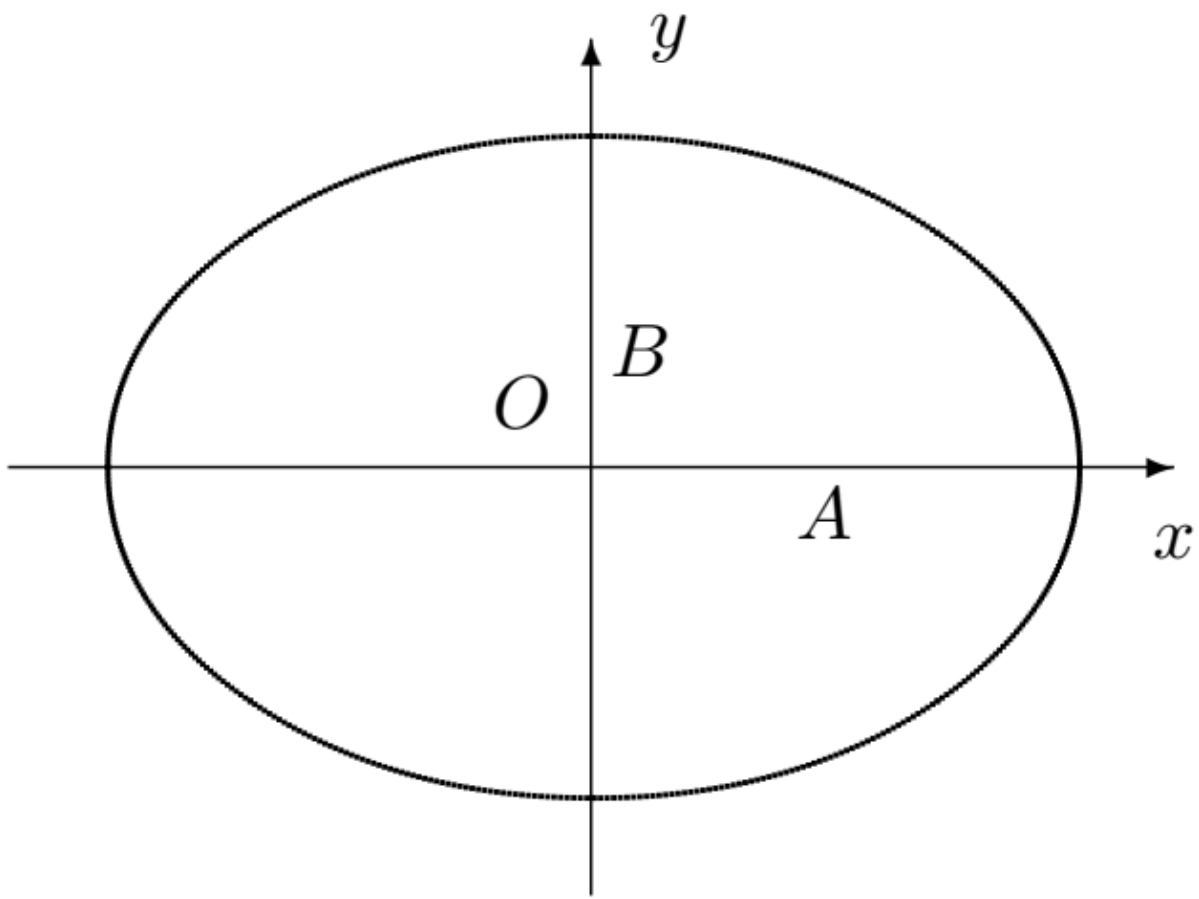
$$\frac{y}{B} = \frac{x}{A}\cos\delta - \operatorname{sen}\delta\sqrt{1 - \frac{x^2}{A^2}},$$

$$\cos\omega_0 t = \frac{x(t)}{A}$$

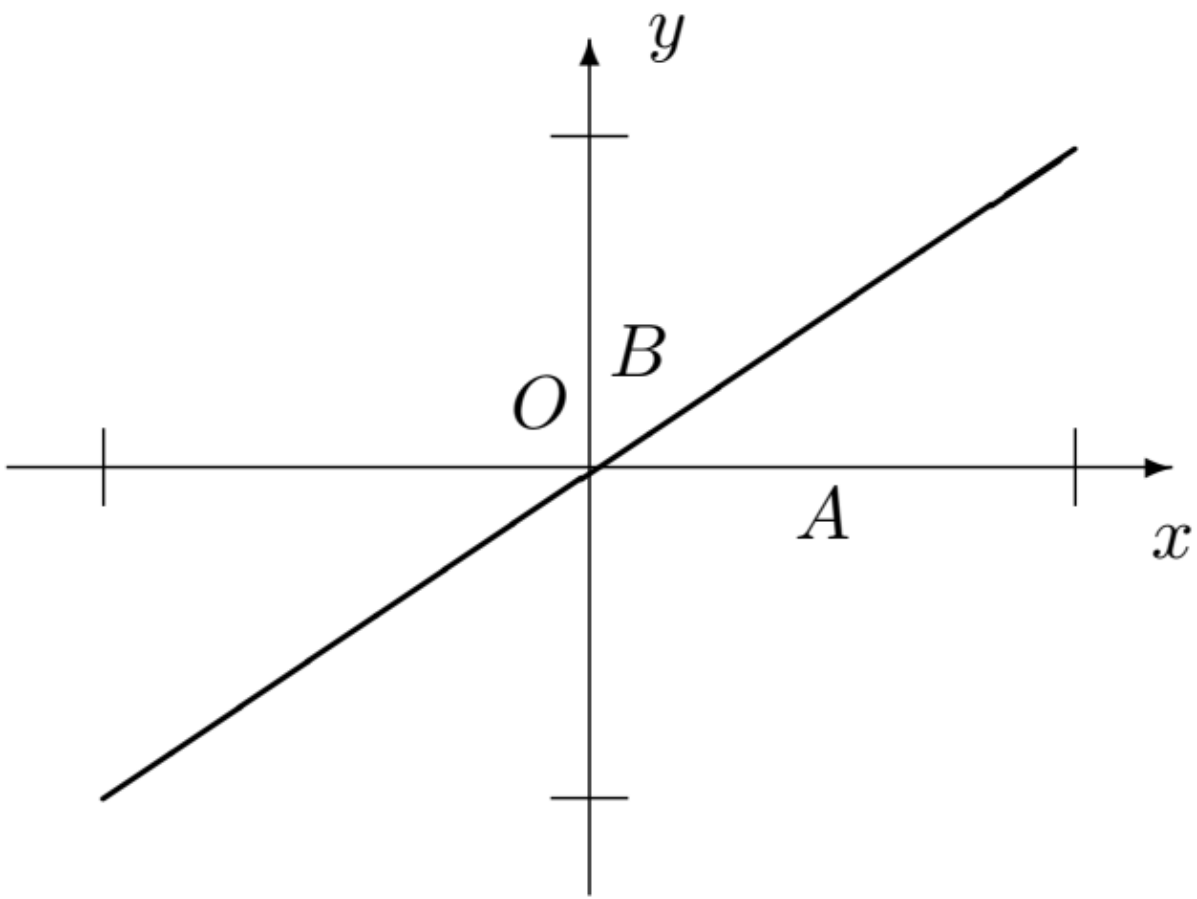
$$\sin\omega_0 t = \sqrt{1 - \cos^2\omega_0 t} = \sqrt{1 - \frac{x^2(t)}{A^2}}$$

$$\frac{x^2}{A^2} + \frac{y^2}{B^2} - \frac{2xy}{AB}\cos\delta = \operatorname{sen}^2\delta.$$

$$\frac{x^2}{A^2} + \frac{y^2}{B^2} - \frac{2xy}{AB}\cos\delta = \operatorname{sen}^2\delta$$



$\delta = \frac{\pi}{2};\ \frac{x^2}{A^2} + \frac{y^2}{B^2} = 1$



$\delta = 0;\ \frac{x}{A} - \frac{y}{B} = 0$

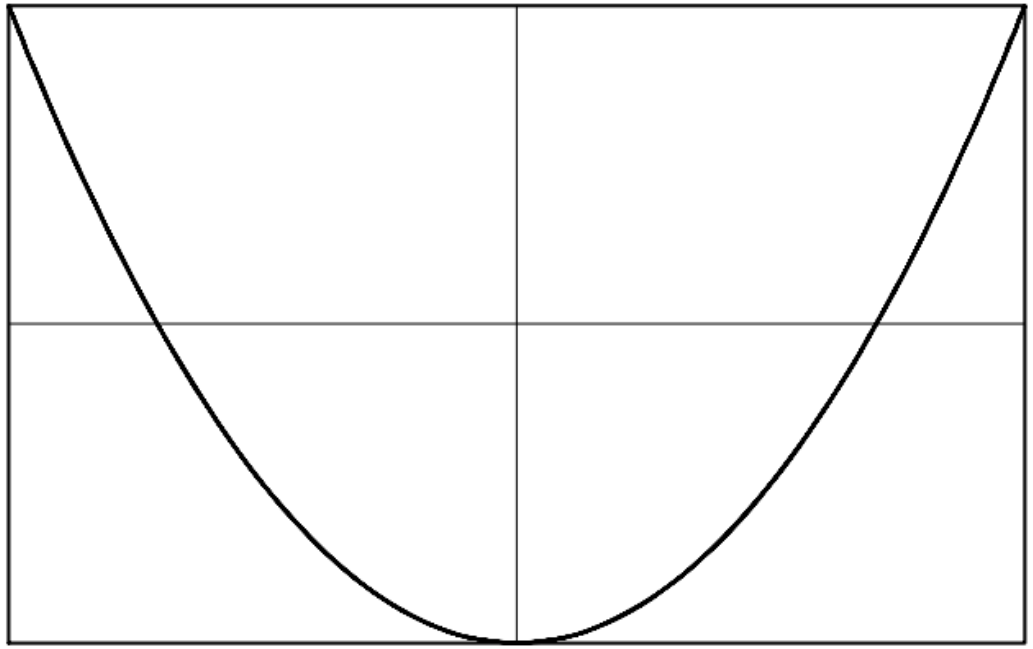
$$\frac{x^2}{A^2} + \frac{y^2}{B^2} - \frac{2xy}{AB}\cos\delta = \operatorname{sen}^2\delta$$

$$\omega_x = \sqrt{k_x/m} \qquad \omega_y = \sqrt{k_y/m}$$

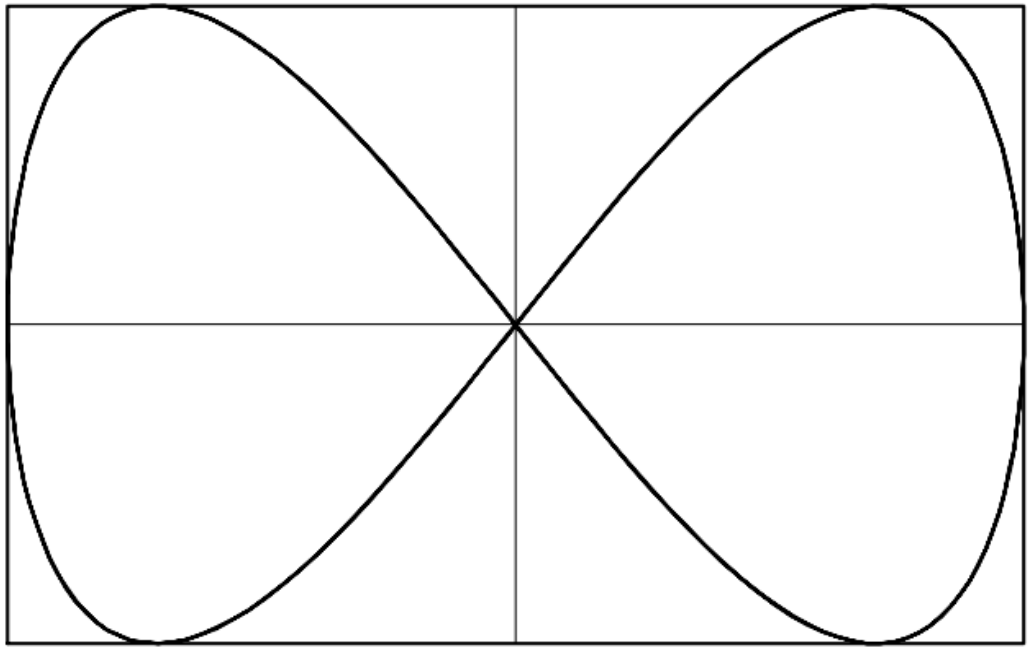
$$x = A\cos(\omega_x t + \alpha), \qquad y = B\cos(\omega_y t + \beta)$$

Essas curvas são fechadas apenas se ω_y/ω_x for uma fração racional

$$x_0 = A, \dot{x}_0 = 0, \qquad y_0 = 0, \dot{y}_0 = -B\omega_0$$



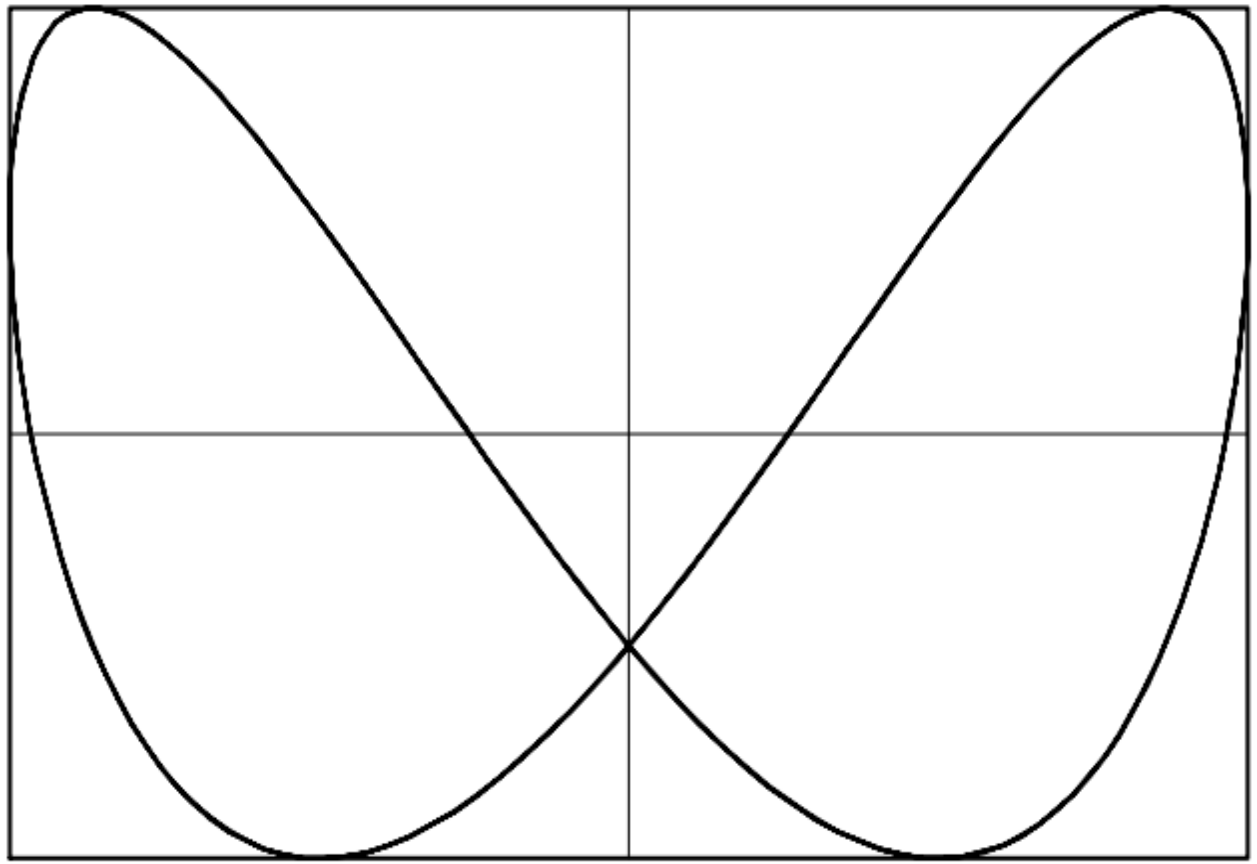
$\omega_y = 2\omega_x;\ \ \delta = 0$



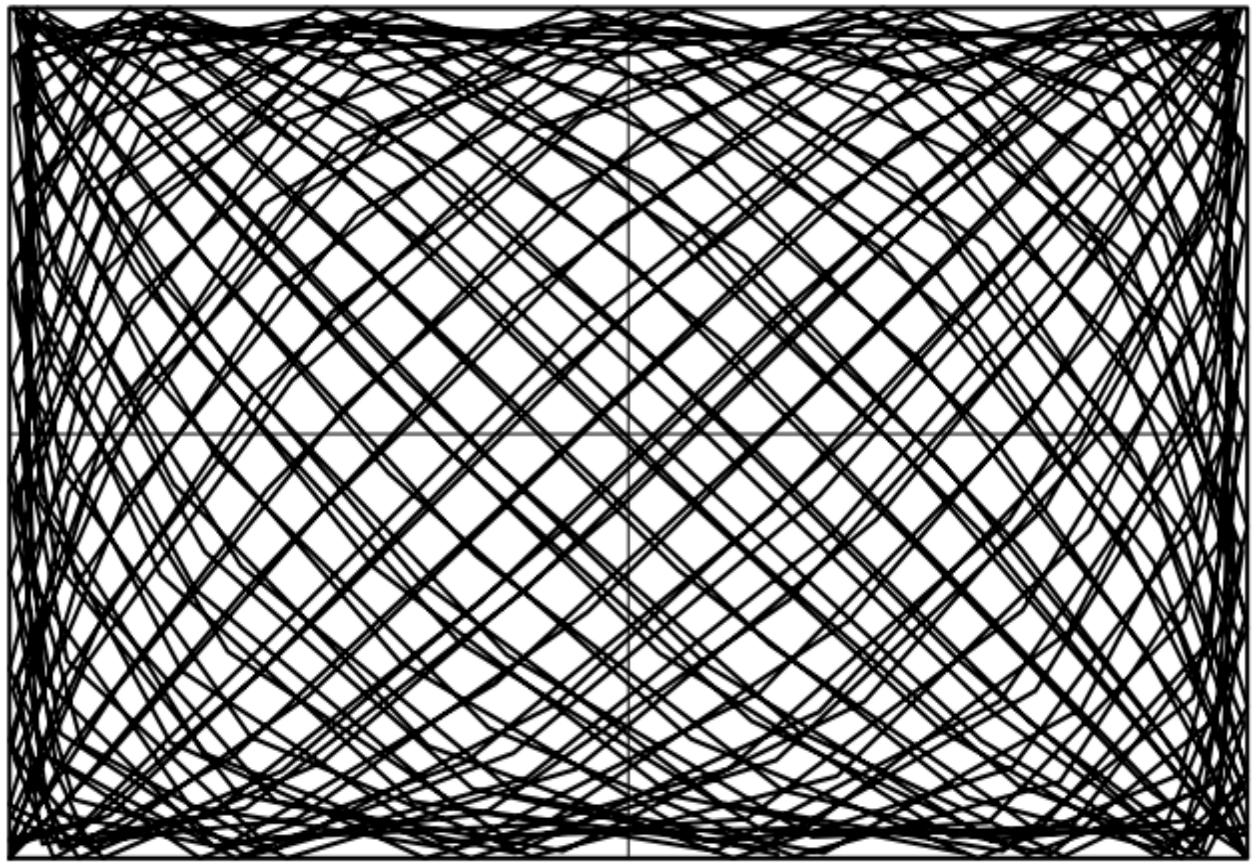
$\omega_y = 2\omega_x;\ \ \delta = \pi/2$

Curvas de Lissajous

$$x_0 = A, \dot{x}_0 = 0, y_0 = B, \dot{y}_0 = 0.$$

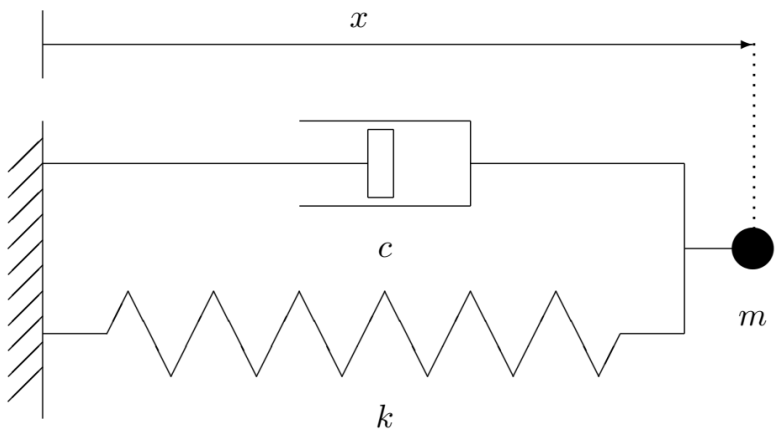


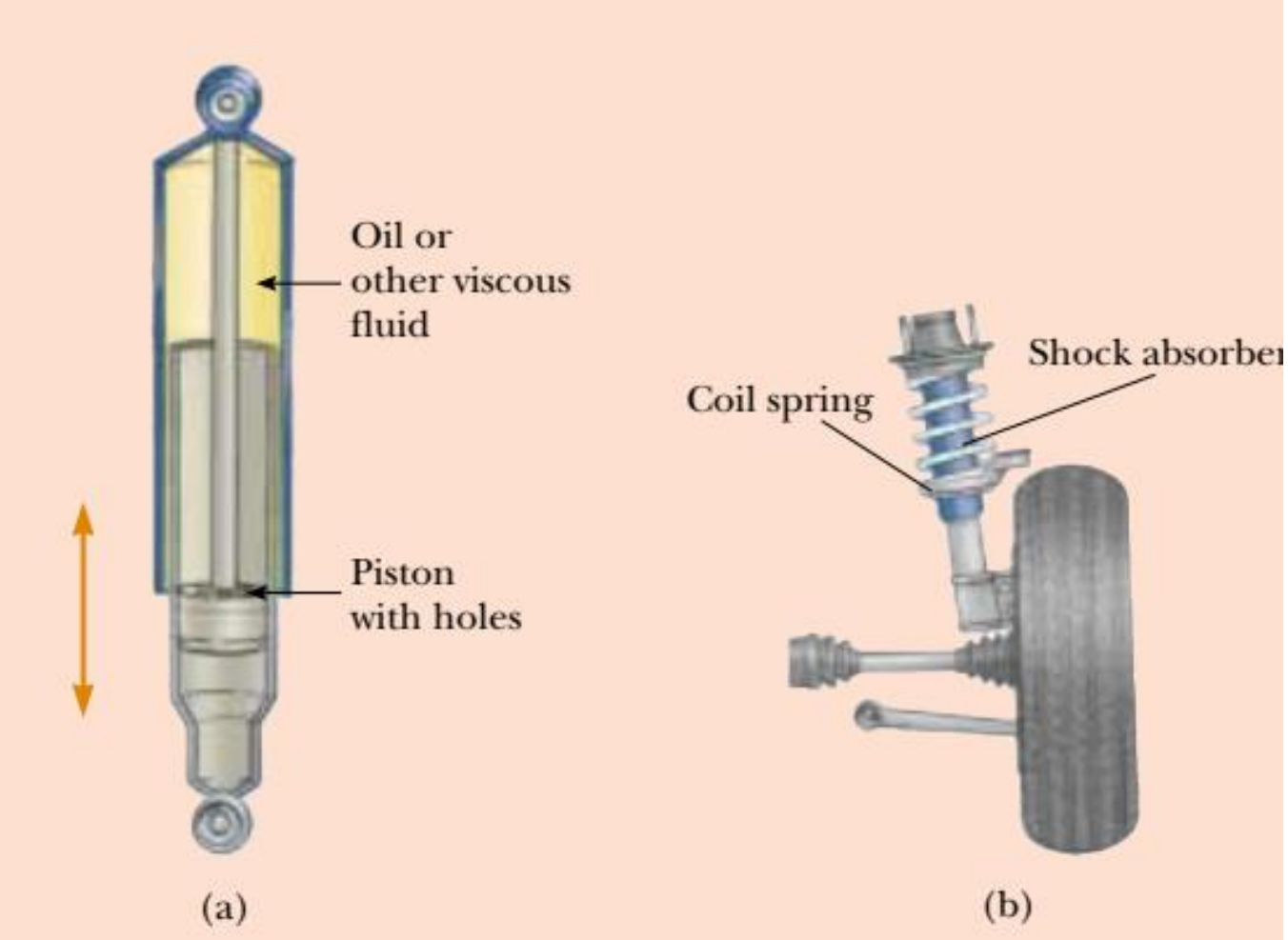
$\omega_y = 2\omega_x;\ \delta = \pi/3$

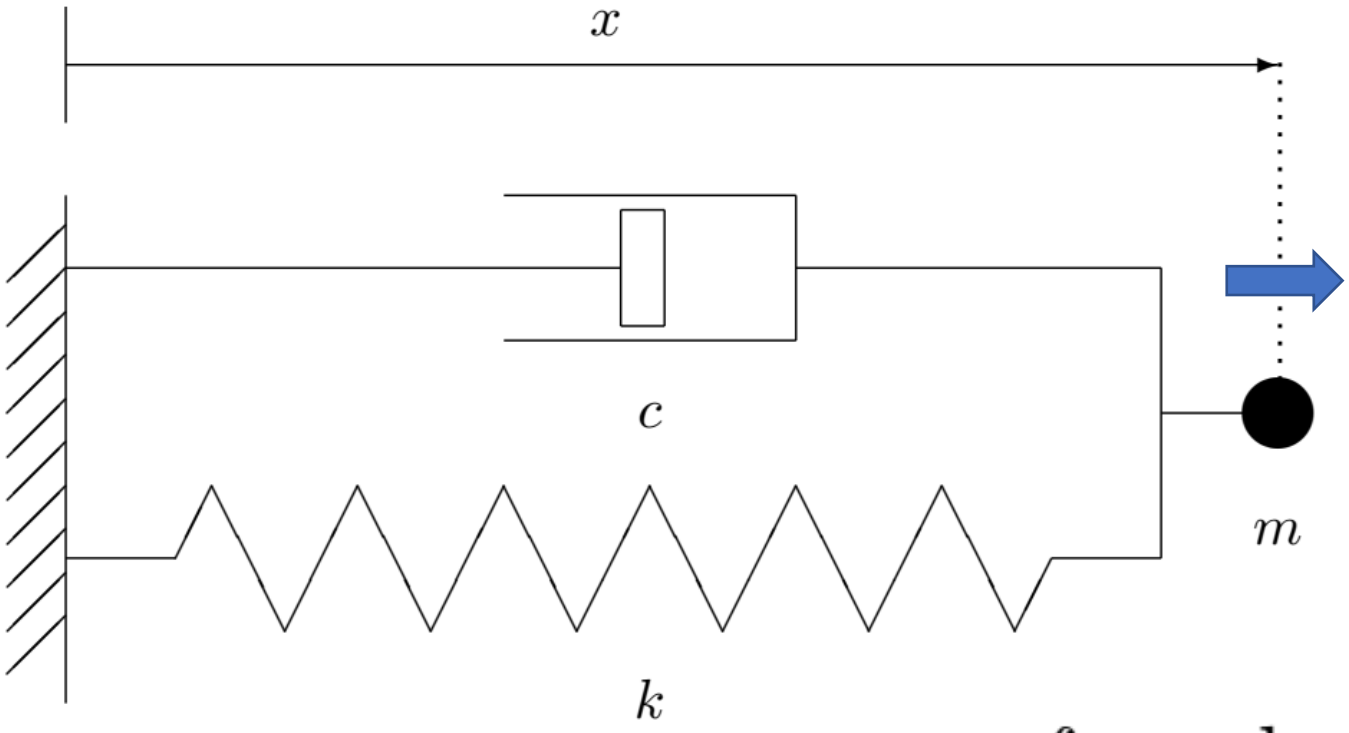


$\omega_y = \sqrt{2}\omega_x;\ \delta = \pi/2$

Oscilações amortecidas







força de retardo $\vec{\mathbf{R}} = -b\,\vec{\mathbf{v}}$

$$W_A = \oint \left(-b\,\frac{\mathrm{d}x}{\mathrm{d}t}\right)\mathrm{d}x = \oint \left(-b\dot{x}^2\right)\mathrm{d}t < 0.$$

$$W_A = \oint -b\frac{dx}{dt}\frac{dx}{dt}\,dt$$

$$\sum F_x = -kx - bv = ma_x \quad \Rightarrow \quad -kx - b\frac{dx}{dt} = m\frac{d^2x}{dt^2}$$

$$\ddot{x} + \frac{b}{m}\dot{x} + \frac{k}{m}x = 0$$

$$\boxed{x = ce^{ht}} \quad \Rightarrow \quad \dot{x} = che^{ht} \quad \Rightarrow \quad \ddot{x} = ch^2e^{ht}$$

$$\ddot{x} + \frac{b}{m}\dot{x} + \frac{K}{m}x = 0$$

$$Ch^2e^{ht} + Ch\frac{b}{m}e^{ht} + C\frac{K}{m}e^{ht} = 0$$

$$Ce^{ht}\left(h^2 + h\frac{b}{m} + \frac{K}{m}\right) = 0$$

$$h^2 + h\frac{b}{m} + \frac{K}{m} = \emptyset$$

$$h = -\frac{b/m \pm \sqrt{(b/m)^2 - 4K/m}}{2}$$

$$h_{1,2} = -\frac{b \pm \sqrt{b^2 - 4Km}}{2m}$$

$$h_1 = \frac{-b + \sqrt{b^2 - 4Km}}{2m} \quad ; \quad h_2 = \frac{-b - \sqrt{b^2 - 4Km}}{2m}$$

$$x(t) = C_1 e^{h_1 t} + C_2 e^{h_2 t}$$

$$x(t) = C_1 e^{\left(\frac{-b + \sqrt{b^2 - 4Km}}{2m}\right)t} + C_2 e^{\left(\frac{-b - \sqrt{b^2 - 4Km}}{2m}\right)t}$$

$$x(t) = c_1 e^{\left(\frac{-b+\sqrt{b^2-4km}}{2m}\right)t} + c_2 e^{\left(\frac{-b-\sqrt{b^2-4km}}{2m}\right)t}$$

criticamente amortecido $\longrightarrow$ $b = \sqrt{4\ \mathrm{km}}$

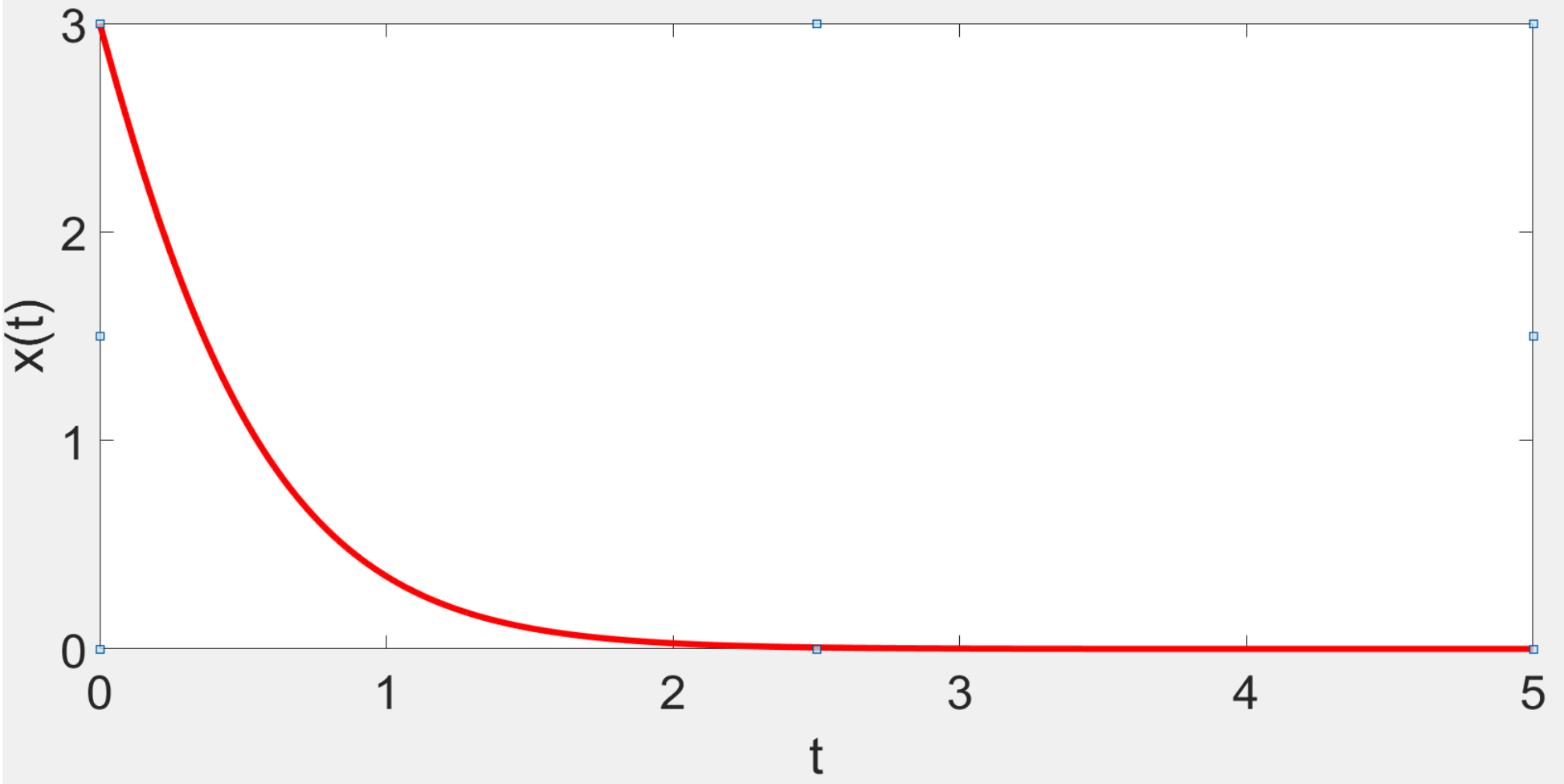
$$\sqrt{b^2-4km} = 0$$

$$x(t) = c_1 e^{\frac{-b}{2m}t} + c_2 e^{\frac{-b}{2m}t} = \underbrace{(c_1+c_2)}_{a_1} e^{\frac{-b}{2m}t}$$

$$x(t) = a_1 e^{\frac{-b}{2m}t}$$

- Pode-se verificar que $a_2 t e^{-\frac{b}{2m}t}$ também é uma solução, portanto a solução geral será uma combinação destes dois:

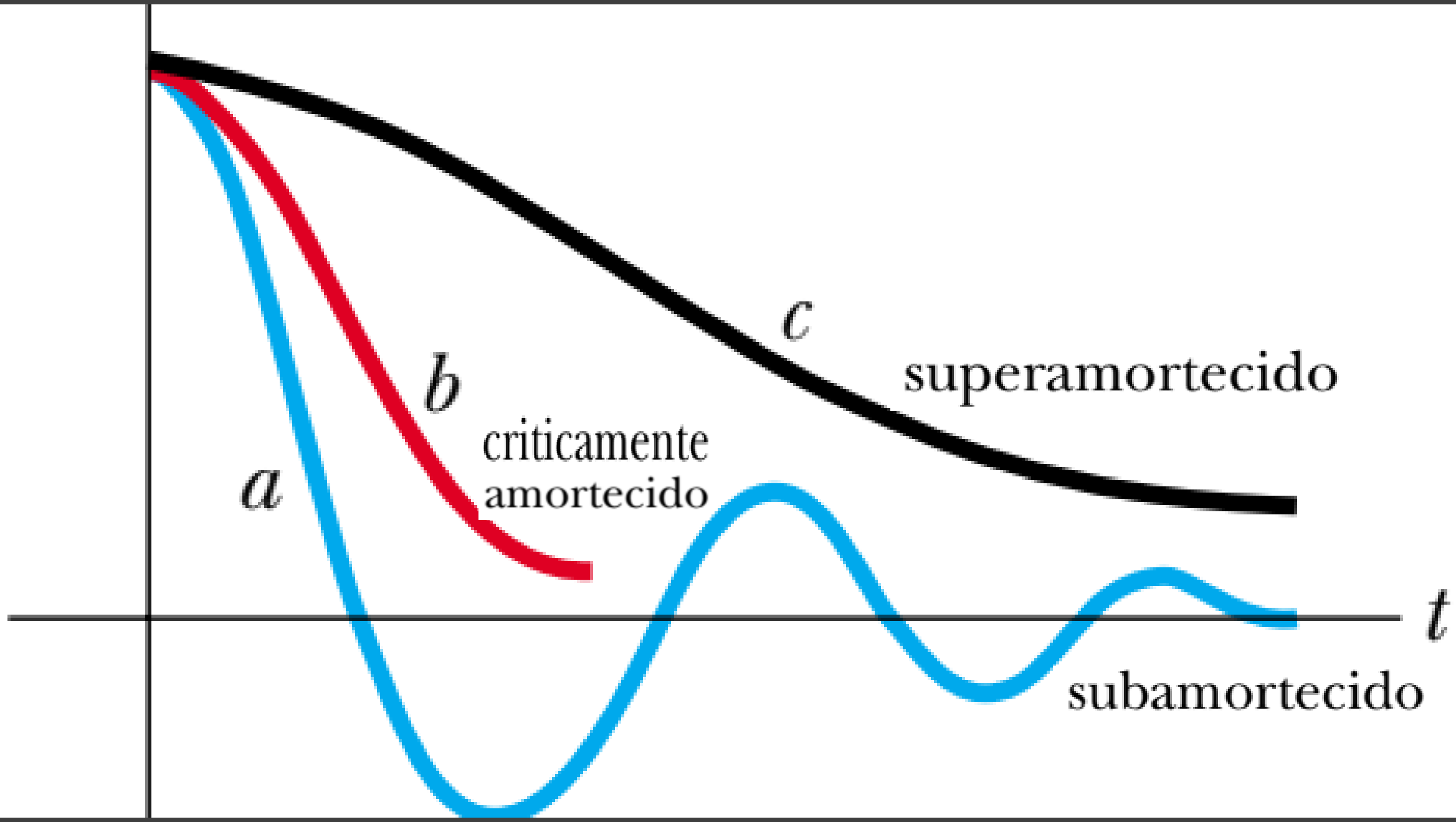
$$x(t) = (a_1 + a_2 t)\, e^{-\frac{b}{2m}t}$$



$$x(t) = c_1 e^{\left(\frac{-b+\sqrt{b^2-4km}}{2m}\right)t} + c_2 e^{\left(\frac{-b-\sqrt{b^2-4km}}{2m}\right)t}$$

superamortecido $\Rightarrow$ $b^2 - 4Km > 0$

$$x(t) = e^{\frac{-bt}{2m}}\left[c_1 e^{\frac{\sqrt{b^2-4Km}}{2m}t} + c_2 e^{-\frac{\sqrt{b^2-4Km}}{2m}t}\right]$$


c
superamortecido
b
criticamente
amortecido
a
t
subamortecido

$$x(t) = c_1 e^{\left(\frac{-b+\sqrt{b^2-4km}}{2m}\right)t} + c_2 e^{\left(\frac{-b-\sqrt{b^2-4km}}{2m}\right)t}$$

subamortecido $\rightarrow$ $b^2 - 4Km < 0$

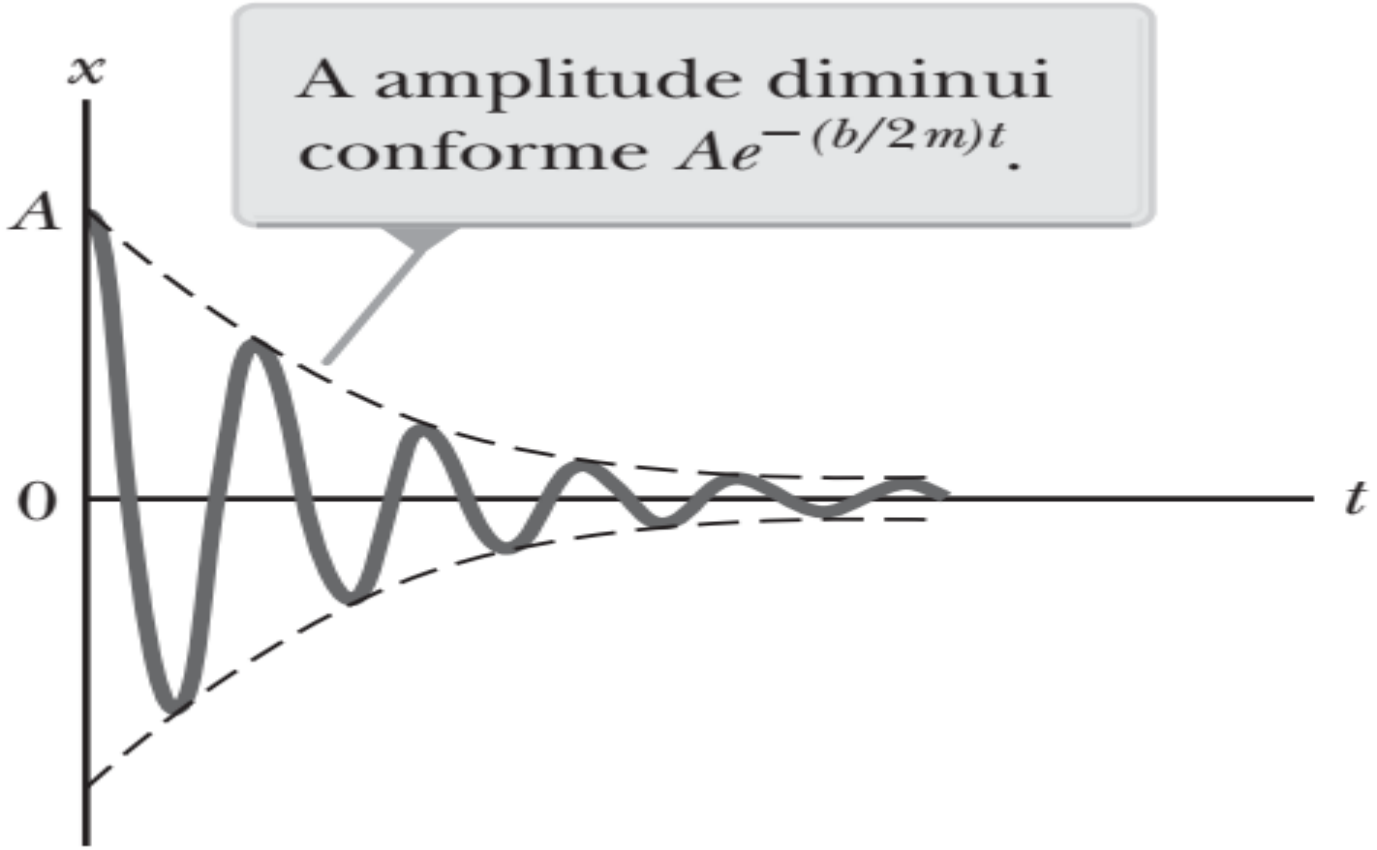
$$\sqrt{b^2 - 4Km} \rightarrow \sqrt{(-1)}\sqrt{4Km - b^2} = i\sqrt{4Km - b^2}$$

$$x(t) = e^{-\frac{bt}{2m}}\left[c_1 e^{\frac{i\sqrt{4Km-b^2}}{2m}t} + c_2 e^{\frac{-i\sqrt{4Km-b^2}}{2m}t}\right]$$

$$e^{\pm i\theta} = \cos\theta \pm i\sin\theta$$

$$x(t) = Ae^{-\frac{bt}{2m}}\sin(\omega t + \varphi)$$

$$\omega = \sqrt{\omega_0{}^2 - \left(\frac{b}{2m}\right)^2} \quad \Rightarrow \quad \omega_0 = \sqrt{k/m}$$



$$\xi \stackrel{\text{def}}{=} \frac{b}{b_{\text{crit}}} = \frac{b}{2m\omega_0}.$$

$$\xi \geq 1 \begin{cases} \textbf{criticamente amortecido} \\ \\ \textbf{superamortecido} \end{cases}$$

Para vibrações estruturais, os valores usuais de

$$\xi = 0{,}02 = 2\,\%$$

$$\ddot{x} + 2\xi\omega_0\dot{x} + \omega_0^2 x = 0,$$

$$x = a\mathrm{e}^{-\xi\omega_0 t}\,\mathrm{sen}(\omega t + \varphi) \qquad \omega \stackrel{\mathrm{def}}{=} \omega_0\sqrt{1-\xi^2}$$

Exemplo

Uma massa m de 400 kg pode deslizar sem atrito sobre um eixo horizontal, fixado por uma mola elástica de k = 10^5 N/m constante a uma base fixa no eixo. Existe também um amortecimento viscoso, que reduz a amplitude da oscilação a uma centésima parte a cada 10 s. É solicitado

O valor da constante de amortecimento *b* e da taxa ξ em relação a do crítico

$$x = a\mathrm{e}^{-\xi\omega_0 t}\,\mathrm{sen}(\omega t + \phi_0).$$

$$\omega_0 = \sqrt{k/m} = 15{,}811388\,\mathrm{rad/s}$$

$$a\mathrm{e}^{-\xi\omega_0\cdot 10} = \frac{a}{100} \quad \Rightarrow \quad \xi = \frac{\ln 100}{10\omega_0} = 0{,}029126 \approx 2{,}9\,\%.$$

$$\boxed{b = 2\xi m\omega_0 = 368{,}4136\,\mathrm{N\cdot s\cdot m^{-1}}}$$

- Supondo que parte de x0 = 0,05 m medido a partir da posição de equilíbrio em repouso, obtenha a equação do movimento e também o valor numérico da posição após 2 s.

$$x = a\mathrm{e}^{-\xi\omega_0 t}\,\mathrm{sen}(\omega t + \varphi)$$

$$\dot{x}(t) = A\left[-\xi\omega_0 e^{-\xi\omega_0 t}\sin(\omega t+\varphi) + e^{-\xi\omega_0 t}(\omega)\cos(\omega t+\varphi)\right]$$

$$t=0 \rightarrow x(0)=0{,}05\,m \rightarrow \dot{x}(0)=0\,m/s$$

$$\left.\begin{aligned} x_0 &= A \operatorname{sen} \phi_0 \\ 0 &= -A\xi\omega_0 \operatorname{sen} \phi_0 + A\omega \cos \phi_0 \end{aligned}\right\} \Rightarrow \left\{\begin{aligned} A &= 0{,}05002122 \\ \phi_0 &= 1{,}541667 = 0{,}490728\pi \end{aligned}\right.$$

$$x(0) = 0{,}05 = A \sin \varphi$$

$$\dot{x}(0) = 0 = -A\xi\omega_0 \sin \varphi + A\omega \cos \varphi$$

$$0 = -0{,}05\,\xi\omega_0 + \left(\frac{0{,}05}{\sin \varphi}\right)\omega \cos \varphi$$

$$\cancel{0{,}05}\,\xi\omega_0 = \cancel{0{,}05}\,\omega \operatorname{ctg} \varphi$$

$$\frac{\xi\omega_0}{\omega} = \operatorname{ctg} \varphi \Rightarrow \operatorname{tg} \varphi = \frac{\omega}{\xi\omega_0} \Rightarrow \varphi = \operatorname{tg}^{-1}\left(\frac{\omega}{\xi\omega_0}\right)$$

$$\varphi = tg^{-1}\left(\frac{\omega}{\xi\omega_0}\right) \longrightarrow \omega = \omega_0\sqrt{1-\xi^2}$$

$$\varphi = tg^{-1}\left(\frac{\cancel{\omega_0}\sqrt{1-\xi^2}}{\xi\,\cancel{\omega_0}}\right) \longrightarrow \varphi = tg^{-1}\left(\sqrt{\frac{1}{\xi^2}-1}\right) \cong 0{,}5\pi \text{ rad}$$

$$0{,}05 = A\sin\varphi$$

$$A = \frac{0{,}05}{\sin\varphi} = 0{,}05002122$$

$$\begin{cases} A = 0{,}05002122 \\ \phi_0 = 1{,}541667 = 0{,}490728\pi \end{cases}$$

$$x(t) = 0{,}05002122\,\mathrm{e}^{(-0{,}4605170\,t)}\ \mathrm{sen}(15{,}80468\,t + 1{,}541667)$$

$$\boxed{x(2) = 0{,}01964561\,\mathrm{m}.}$$